AdBuster PRO – TV Volume Stabilizer

Version 2.0.0-PRO (2026) A fully autonomous, ML-powered TV volume stabilizer with Broadlink IR control, real-time audio analysis, CEPA Human logic, and optional PRO features.

What AdBuster Does

AdBuster listens to your TV audio through your computer's microphone, analyzes the sound in real time, and automatically adjusts the TV volume using a Broadlink IR blaster.

It reacts to:

• sudden loud spikes

• commercials

• music transitions

• quiet dialogue

• nighttime vs daytime

• user manual changes

• longterm audio patterns

All without touching the remote.

Machine Learning

AdBuster includes two independent ML engines:

ML1 – Lightweight Online Model (always active)

• Learns continuously every second

• Saves training data automatically

• Stores model parameters in model.pkl

• Improves stability and reaction accuracy over time

ML PRO – Advanced Classifier (optional PRO feature)

• Uses RMS history (50 samples)

• Extracts features (mean, std, delta, range)

• Classifies audio into:

• AD (commercials)

• MUSIC

• TALK
• NORMAL
• Trains a RandomForest model
• Saves model to model_deep.pkl
• Allows user-driven training via GUI
Both ML engines work offline, store data locally, and improve over time.

PRO Mode & License Keys

AdBuster PRO includes an optional unlockable module with advanced ML features.

How PRO Unlock Works

The app uses a **fully offline license system**:

- No internet activation
- No server communication
- No online database
- No tracking

The app checks keys **locally**, using:

1. **MASTER KEY** (always valid – for the developer)
2. **USER KEYS** from `keys.json`
3. **Algorithmic Keys** (validated by checksum)
4. **Daily attempt limit** (3 tries per day)

How users unlock PRO

1. Download the free ZIP
2. Run the app (PRO is locked)
3. Support the project (e.g., buy a coffee)
4. Receive a license key by email
5. Enter the key in the app
6. PRO unlocks instantly, offline

You do **not** need a server, database, or account system. The app validates keys by itself.

Supported Broadlink Devices (IR Models)

AdBuster PRO works with **any Broadlink device that supports IR sending**. It is **not limited** to RM Mini 3.

✓Fully Supported IR Models

- Broadlink RM Mini 3

- Broadlink RM Mini 4
- Broadlink RM4C Mini
- Broadlink RM4 Mini
- Broadlink RM Pro / RM Pro+
- Broadlink RM4 Pro
- Broadlink RM4 TV Mate
- Broadlink RM3 Black Bean

If the device can:

- enter IR learning mode
- capture IR
- send IR

…it will work with AdBuster.

❌ Not Supported (No IR)

- Smart plugs (SP series)
- Sensors (HTS2, etc.)
- Wi-Fi switches
- Curtain motors
- RF-only devices

How detection works

AdBuster uses:

- `broadlink.discover()` → finds all Broadlink devices
- IP + MAC → selects the correct one
- `device.enter_learning()` → learns your remote
- `device.send_data()` → sends IR commands

Key Features

- Real-time audio analysis
- Automatic volume correction
- Commercial detection
- Music/Talk mode detection
- CEPA Human behavior simulation
- ML1 + ML PRO hybrid engine
- Broadlink IR control

- Azure-styled GUI
- Calibration mode
- Daytime threshold profiles
- Telemetry logs (A/B rotation)
- Offline license system
- RMS history
- Spike detection
- Manual override detection
- Watchdog auto-reconnect

Installation (EXE version)

1. Download the ZIP package.
2. Extract all files to any folder.
3. Create a desktop shortcut to **Start.bat**.
4. Right-click the shortcut → *Properties* → *Change Icon…* → select the **included icon file** (`AdBuster.ico`).
5. Launch the application using the shortcut.
    - The launcher automatically starts **AdBuster.exe**, **Aduster.exe**, and **VolMaster.exe**.
    - All other .exe files remain in the folder and operate as internal cooperating modules.
6. No Python or additional software is required.

Broadlink Requirement

AdBuster requires a **Broadlink IR device** to operate. Before first use, complete the device setup in the **Broadlink app**. Once configured, AdBuster will automatically communicate with the Broadlink device to control your TV.

Flask Server Configuration (IP, MAC, Port 5000)

To enable communication with the Broadlink device, the Flask server must be configured with:

- **Broadlink IP address** (e.g., `192.168.1.25`)
- **Broadlink MAC address in UPPERCASE, digits and letters only, without any separators**, for example: `47A1B2CDE3F4` not `47:A1:B2:CD:E3:F4` not `47-a1-b2-cd-e3-f4` not `47a1b2cde3f4` (lowercase)
- **Port 5000**, which is the default communication port used by the Flask server.

The MAC address must be formatted exactly as shown above. If it contains colons, dashes, lowercase letters, or any special characters, the server will not detect the device correctly.

How the Application Works

- **Start.bat** is the only correct entry point for launching the system.
- It automatically starts three cooperating modules:
  - `AdBuster.exe`
  - `Aduster.exe`
  - `VolMaster.exe`
- These modules communicate with each other and must remain in the same folder.
- The user should never run these executables manually.
- The Flask server listens on **port 5000** and handles Broadlink communication.

Optional

If you use Broadlink IR, complete the initial setup once. After that, the app works fully offline.

Support Project

If you enjoy AdBuster and want to unlock PRO features, you can support the project (e.g., buy a coffee). You will receive a **license key by email**, which unlocks PRO instantly.

-Thank you - AdBuster Team